



Mean–variance portfolio optimization with deep learning based-forecasts for cointegrated stocks

Juan Du

School of Finance, Shandong University of Finance and Economics, Jinan, China

ARTICLE INFO

Keywords:

Mean–variance portfolio optimization
Pairs trading
Mean-reverting spread prediction
Stationary portfolio
Attention-based LSTM network

ABSTRACT

Most mean–variance (MV) models construct a portfolio based on nonstationary stocks. This study presents a new MV model constructed using stationary portfolios composed of cointegrated stocks. The expected return of this new model is predicted by using machine learning models, such as support vector machine, random forest, and attention-based long short-term memory (LSTM) network. The proposed model is evaluated using data on stocks in the CSI 300 and the S&P 500, with 42 features over 8 years from May 4, 2012 to August 4, 2020. The empirical results show that the portfolio constructed based on the stationary portfolios in both the Chinese and the US stock markets delivers significant profits. Further, the attention-based LSTM network can more accurately model the spread return using technical indicators, financial investment information, and lagged returns, and can successfully select pairs of cointegrated stocks for constructing a more profitable MV portfolio, than can conventional machine learning models. Using the attention-based LSTM to predict 20-day return movement results in model accuracy of up to 92.59% for the CSI 300 and 88.52% for the S&P 500, and a corresponding Sharpe ratio of 9.31 and 2.77, respectively.

1. Introduction

Predicting stock market return is a fundamental problem in asset pricing and portfolio management. The nonlinear and nonstationary properties of the financial time series pose major challenges for accurate forecasting. Machine learning models hold the promise of overcoming these challenges, for these models can capture the nonlinear effect of predictive variables on return forecasting and construct highly profitable portfolios, compared with the conventional linear models (Gu et al., 2020; Heaton et al., 2017; Wu et al., 2021).

The literature has presented evidence of the co-movement of individual stocks within national markets in the US, Japan, and Taiwan (Green & Hwang, 2009; Greenwood, 2008; Huang et al., 2019). Furthermore, Barberis et al. (2005) and Greenwood (2008) show that the co-movement of stocks in the stock market index is stronger than that of stocks not in that index. All these empirical studies show that the stationary and cointegration analysis of stocks that compose the stock market indices in different countries is necessary. Several theoretical and empirical studies have focused on the mean–variance (MV) portfolio selection of cointegrated assets have been developed (Chiu & Wong, 2011, 2012; Macedo et al., 2017). Nevertheless, few machine learning studies in the field of stock return prediction and portfolio construction pay attention to stationary portfolios.

Pairs trading is a quantitative investment strategy based on the notion of two stock prices “co-moving” with each other. If the two prices

diverge, a long–short position can be used to profit from the expected reconvergence of the prices. Cointegration-based pairs trading relies on the construction of mean-reverting spreads based on the long-term cointegration relationship of paired stocks and generates profits from rapid convergence of the spreads. Recent studies show that conventional machine learning models can be successfully applied in statistical arbitrage (Huck, 2019; Krauss et al., 2017). In particular, the long short-term memory (LSTM) network can identify pairs trading opportunities among cointegrated stocks composing the S&P 500 (Flori & Regoli, 2021). Motivated by the co-movement evidence and the strong return shown in the discussed studies, this study employs machine learning models and cointegration-based pairs trading to construct stationary portfolios between the co-moving stocks in the S&P 500 and the CSI 300.

The MV model proposed by Markowitz (1952) is the first mathematical formalization of the investment diversification concept. Because theoretical studies have commonly revealed that MV allocations are very sensitive to small forecast errors in the means and covariances (Chopra et al., 1993; Goldfarb & Iyengar, 2003), a growing body of literature has focused on extending Markowitz's MV model by enhancing the estimate accuracy of the expected returns using machine learning models. Indeed, these extended models are superior to traditional MV models (without stock prediction) in terms of returns

E-mail address: juandu@sdufe.edu.cn.

<https://doi.org/10.1016/j.eswa.2022.117005>

Received 2 September 2021; Received in revised form 22 January 2022; Accepted 26 March 2022

Available online 12 April 2022

0957-4174/© 2022 Elsevier Ltd. All rights reserved.

and risks (Chen et al., 2021; Freitas et al., 2009; Krauss et al., 2017; Ma et al., 2021; Paiva et al., 2019; Wang et al., 2020; Yu et al., 2020). The literature covers various models used for forecasting stock returns, such as autoregressive moving reference neural network, autoregressive integrated moving average, deep multilayer perceptron, LSTM network, support vector machine (SVM), convolutional neural network, random forest, and extreme gradient boosting. All these related studies show that among the models considered, machine learning models, such as neural network, SVM, LSTM networks, random forest, and extreme gradient boosting, outperform non-machine learning models in predicting preselected stock returns, but most of these studies ignore the stationary analysis in the preselection process. This lack of research attention will lead to inaccurate stock return predictions and will thus negatively affect MV allocations.

Given the existing literature gaps, the contribution of this study is threefold. First, to our knowledge, this study is unique in terms of portfolio construction using both machine learning and quantitative investment techniques. Unlike prior studies that combine MV models with machine learning-based predictions, the new model proposed in this study incorporates stationary analysis and machine learning predictive models into MV portfolio optimization. The final constructed portfolio has three advantages. The portfolio is stationary for it is composed of stationary portfolios. It is a hedge portfolio for it is composed of the top cointegrated pairs portfolios. It is a diversified portfolio under the rule of MV optimization.

Second, this is the first study to introduce a holding period and a trading threshold in an MV portfolio optimization model, which are essential variables for a profitable pairs trading strategy. The longer the holding period, the more likely it is that the return of the spread of paired stocks will revert to the long-term mean. The trading threshold balances the trading profits in each trade and the total transaction cost determined by the number of trade occurrences.

Third, this study is the first to use the attention-based LSTM network (A-LSTM network) to integrate short-term signals (technical indicators and lagged returns) and long-term signals (financial ratios) with different weights to predict an accurate, robust spread of paired stocks. The use of the A-LSTM network further improved the predictability and profitability of the proposed model compared with conventional machine learning models.

Thus, the purpose of this study is to establish an MV portfolio optimization model using stationary pairs trading portfolios that can be predicted by A-LSTM networks. The model is evaluated on both the Chinese and the US stock markets. Specifically, this model involves three main phases related to the pairs trading portfolio: construction, return prediction, and allocation. In the first phase, all binary cointegrated stock pairs are selected to compose pairs trading portfolios based on the Engle–Granger cointegration test and the Johansen test, which convert each of two nonstationary time series stocks into one stationary portfolio. In the second phase, for all selected stock pairs, machine learning models are trained with the technical indicators and the financial information of these cointegrated pairs of stocks to predict the return of the constructed pairs trading portfolios over the holding periods, pair by pair. The pairs trading portfolios with the highest return prediction accuracies are selected for different holding periods. In the third phase, the pairs trading portfolios after the two-step portfolio selection are utilized to construct the MV portfolio. The MV model for constructing the portfolio includes the parameters to measure the investment horizon and return expectation by holding period and trading threshold.

The remainder of the paper is organized as follows. Section 2 describes the structure of the spread prediction-based portfolio optimization model. Section 3 explains the details of the experiments performed. Section 4 provides experimental results on the model predictability and profitability. Section 5 presents conclusions.

2. Spread prediction-based portfolio optimization model

This section presents the components of the spread prediction-based portfolio optimization model, which are cointegration-based pairs trading, the A-LSTM network, and the MV portfolio optimization model. The theories and models for constructing a stationary and optimal MV portfolio are provided.

2.1. Cointegration-based pairs trading

The trading mechanism of cointegration pairs trading has been developed by [Vidyamurthy \(2004\)](#) through taking the arbitrage opportunity of temporary anomalies between prices of related assets that have a long-run equilibrium. Mathematically, if the linear combination of two stocks is unit-root stationary, it implies that the long-term forecasts of the linear combination converge to a mean value. Suppose two stocks (A and B) are cointegrated, the spread established in Eq. (1), is a mean-reverting time series.

$$sr = r_A - \gamma r_B \quad (1)$$

where r_A and r_B are the log return of stock A and B.¹ The spread sr is formed by buying one share of stock A and short selling γ shares of stock B.

[Huck \(2010\)](#) define the h-day-ahead forecast for pairs trading and construct the spread by longing one share of stock A and shorting one share of stock B with an h period forecast horizon. [Flori and Regoli \(2021\)](#) define the spread movement within a h-day horizon for pairs trading and construct the spread by buying one share of stock A and shorting γ shares of stock B. In the proposed model, the holding period of the pairs trading portfolio is considered since the longer the holding period, the more likely it is that the return of the spread will revert to its long-term mean. To show a general case, the spread at time t is established by buying a share in stock A, short selling γ^H shares in stock B, and then holding this portfolio in the following H days. The H-day ahead pairs trading portfolio's spread ($sr^{t+H|t}$) is

$$sr^{t+H|t} = r_A^{t+H|t} - \gamma^H r_B^{t+H|t} \quad (2)$$

where the log return ($r_i^{t+H|t}$) for stock i is $\ln(S_i^{t+H}/S_i^t)$, $i = A, B$. H is the holding period of a pair $sr^{t+H|t}$, which could be up to 20 trading days ($H = 1, 2, 3, 4, 5, 10, 15, 20$) following [Huck \(2010\)](#).

Each pair of stocks is selected according to the Engle–Granger cointegration test and the Johansen test. Following [Flori and Regoli \(2021\)](#), the hedging ratio (γ) for each spread is estimated through ordinary least squares regression. To maintain a stationary spread over the entire sample period, the previous year's historical log returns of every two stocks are utilized to compute the hedging ratio, which is dynamically changed every year.

2.2. Architecture of attention-based LSTM network

The end-to-end neural network model for extracting features using various recurrent neural network and convolutional neural network neurons on different time series samples that [Long et al. \(2019\)](#) has developed separates mixed-frequency time series sequences for training. This paper offers an end-to-end neural network, the A-LSTM model, to integrate short- and long-term signals with different weights to predict an accurate, robust spread. The structure of this attention-based model is shown in [Fig. 1](#).

¹ Since the magnitudes of simple return and log return are small, log return is used for the convenience of calculating the cumulative returns over the holding period.

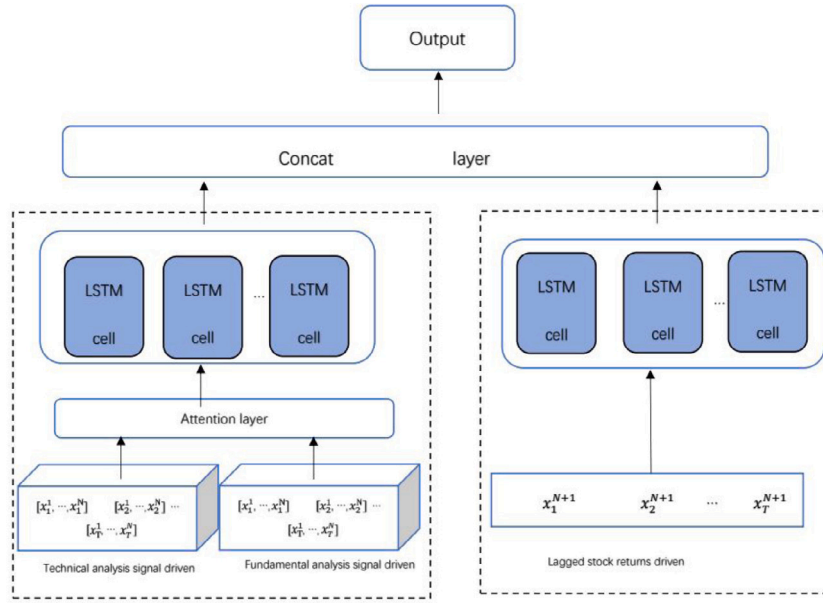


Fig. 1. Structure of attention-based LSTM framework.

The soft attention mechanism that allocates different weights to technical and fundamental analysis signals is implemented using the following Eqs. (3) to (5).

$$s_t^i = \tanh(w_x x_t^i + b_x) \quad (3)$$

$$\alpha_t^i = \frac{\exp(s_t^i)}{\sum_{i=1}^N \exp(s_t^i)} \quad (4)$$

$$c_t = \sum_{i=1}^N \alpha_t^i x_t^i \quad (5)$$

where output c_t of the attention model is weighted and averaged with input information x_t^i consisting of all technical and fundamental indicators of stocks A and B. Subsequently, the N multi-variate time series c and the time series x^{N+1} for the lagged spread returns are trained by LSTM cells simultaneously, while the lagged returns-driven data are encoded by other LSTM cells separately.

LSTM is used to predict the spread return y_{t+1} at day $t+1$ based on information of c and x^{N+1} for the $t-m$ to t period.

The inner structure of LSTM units is composed of a forget gate (f_t), an external input gate (i_t), a candidate vector ($\hat{C}_t$), an internal state (C_t), an output gate (O_t), and the hidden layer output (h_t) to maintain and update the state of memory cells, as in Eqs. (6) to (11). The final predicted value ($\hat{y}$) is obtained by Eq. (12).

$$f_t = \sigma(W_f \cdot [h_{t-1}, c_t] + b_f) \quad (6)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, c_t] + b_i) \quad (7)$$

$$\hat{C}_t = \tanh(W_C \cdot [h_{t-1}, c_t] + b_C) \quad (8)$$

$$C_t = f_t * C_{t-1} + i_t * \hat{C}_t \quad (9)$$

$$O_t = \sigma(W_O \cdot [h_{t-1}, c_t] + b_O) \quad (10)$$

$$h_t = O_t * \tanh(C_t) \quad (11)$$

$$\hat{y} = \text{softmax}(W_y h_t + b_y) \quad (12)$$

where each memory cell in the gate has the corresponding weight matrix W and bias b .

All parameters used in the A-LSTM model is estimated using a standard least squares, or “12” objective function:

$$\text{loss} = \frac{1}{2n} \sum_{t=1}^n (y_t - \hat{y}_t)^2 + \frac{\lambda}{2} \|P\|^2 \quad (13)$$

where y_t and $\hat{y}_t$ denote the actual and predicted return, n is the number of days in the test set, P denotes the set of all trainable parameters (W and b), and λ is the regularization weight.

2.3. Mean–variance with forecasting (MVF) model

So far, the portfolios selected for use in portfolio optimization satisfy the following two conditions: According to the Engle–Granger test and the Johansen test, all pairs trading portfolios (or spreads) are mean-reverting. Moreover, robust predictions of the stationary spreads can be captured by the A-LSTM model using lagged spreads, technical indicators, and financial and economic variables.

Theoretical studies show that MV allocations are very sensitive to small forecast errors in the means and covariances (Chopra et al., 1993; Goldfarb & Iyengar, 2003). Because machine learning models have a strong nonlinear fitting ability in predicting stock returns (Chen & Hao, 2017; Chong et al., 2017; Kara et al., 2011; Patel et al., 2015a; Weng et al., 2018), machine learning models have been widely used in estimating the expected return of Markowitz’s MV model (Freitas et al., 2009; Ma et al., 2021; Paiva et al., 2019; Wang et al., 2020; Yu et al., 2020).

Following the model framework presented by Yu et al. (2020) and Ma et al. (2021) for the asset portfolio optimization model, the proposed spread prediction-based portfolio optimization model utilizes the predicted spread returns as the expected returns of each spread. Since most of the estimation risk in optimal portfolios is due to errors in the estimates of expected returns, we follow Ceria and Stubbs (2006), Gregory et al. (2011), Zymler et al. (2011), and Fliege and Werner (2014) to construct a robust portfolio that incorporates estimation errors. Moreover, threshold (M) of the predicted spread returns that triggers the investment decision is considered as a constraint of the model. The profitability of MV model is affected by M as a smaller M is more likely to generate a higher total return but a higher total transaction cost as the result of more frequent trading. The MVF model aims to trade-off between maximizing returns and minimizing risks, which is expressed through a multi-objective optimization formula:

$$\min \sum_{i,j=1}^N w_i w_j \sigma_{ij} - \sum_{i=1}^N w_i \hat{y}_i - \sum_{i=1}^N w_i \bar{e}_i \quad (14)$$

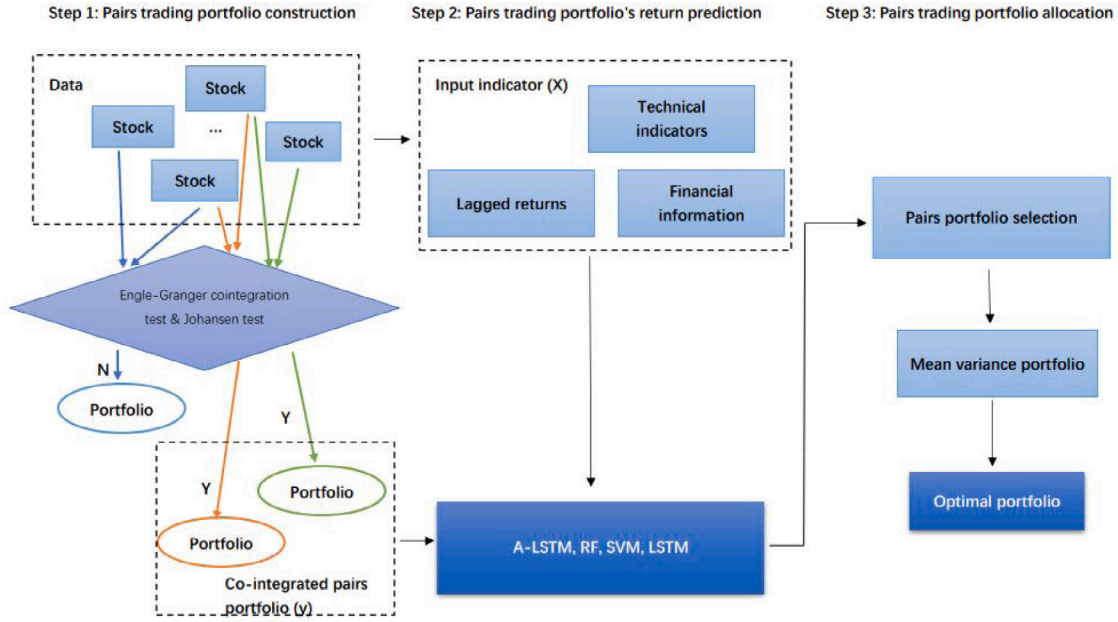


Fig. 2. Overall flowchart of MV model with forecasting.

Subject to

$$\sum_i^N w_i = 1 \quad (15)$$

$$\sum_i^N w_i \hat{y}_i \geq M \quad (16)$$

$$0 \leq w_i \leq 1, i = 1, 2, \dots, N \quad (17)$$

, where

$$\bar{e}_i = y_i - \hat{y}_i \quad (18)$$

where w_i means the weight of the i th spread for the portfolio, N is the number of spread for MV portfolio construction, y_i and $\hat{y}_i$ denote the actual and predicted return of the i th spread, σ_{ij} is the covariance of spreads i and j . The error term $\bar{e}_i$ at time t measures the forecasting error of the spread estimated through the machine learning models. M is the trading threshold for rebalancing the portfolio.

3. Experimental design

This section demonstrates the empirical experiments for implementing the MVF model on the cointegrated stocks using real data from both the Chinese and the US stock markets.

3.1. Data

To verify the effectiveness of the proposed MVF model, data on the constituent stocks of the CSI 300 Index are collected. Since the number of all possible combinations between every pair of stocks that compose the CSI 300 Index is huge, it is difficult to check whether all pairs of the 300 stocks are stationary. In the case of individual investors, it is realistic to consider that they hold less than 20 stocks and that doing so helps them maintain better returns with the same risk level. Therefore, 20 stocks with enough historical data are randomly selected from the CSI 300 Index and are used for pair selection and portfolio information. Their stock tickers are 002179, 002304, 600009, 600011, 600015, 600016, 600028, 600085, 600111, 600519, 600660, 601288, 601328, 601398, 601601, 601607, 601668, 601800, 601898, and 601988.

Furthermore, to avoid sample selection bias, 20 stocks that compose the S&P 500 index – AAPL, AMT, AMZN, AZO, DIS, DVA, EOG, HSY,

INTC, MCK, NVDA, OXY, DEP, RL, SBUX, TXN, VZ, WAB, WMT, and WYNN – are incorporated in the experimental performance check.

The main analysis of the model and trading performance is presented in Sections 4.1 to Section 4.4, which is based on data on China. Section 4.5 examines the most important model performance of the US samples.

3.2. Input indicator

All price records and financial data of Chinese stocks are extracted from the TuShare financial data interface (tushare.org) from May 4, 2012 to August 4, 2020. The price information and financial data for US stocks during the same period are downloaded from the WIND database. The Shanghai interbank offer rate, that is, SHIBOR, is used to approximate the risk-free rate over this period (Hilliard & Zhang, 2015). SHIBOR is available from the website of the China Foreign Exchange Trading Center (CFETCS.org). The US risk-free interest rate is taken from the website of the US Department of the Treasury.

Input indicators, such as technical indicators and firm characters, are chosen as features to train the attention-based LSTM model. Table 1 lists the technical indicators commonly used in prior studies to predict the asset return (Conrad & Kaul, 1998; Han et al., 2013; Kim & Enke, 2016; Nam et al., 2005; Neely et al., 2014) and to categorize the technical indicators into five categories: trend, mean reversion, relative strength, volume, and momentum. The technical analysis is important for the changes in supply and demand for a security can be determined through analyzing market information in the short term. Following Patel et al. (2015a, 2015b), and Alonso-Monsalve et al. (2020), 14 technical indicators are investigated as the features of machine learning models.

The fundamental variables that likely influence future stock returns, and can be used to evaluate a security's intrinsic value from the perspective of share price rationality, profitability, efficiency, growth, leverage, and liquidity, are described in Table 2, in line with Fama and French (2008), Hjalmarsson (2010), Jiang (2021), Lewellen (2004, 2004), and Silva et al. (2015).

Moreover, the technical indicators are updated daily, whereas the fundamental indicators of Chinese stocks are revealed in quarterly and annual financial reports. The technical indicators are utilized to capture the price trend patterns during short periods, and the fundamental

Table 1
Technical indicator data as predictor attributes.

Attribute	Variable	Description
Trend	v1	Simple moving average (30) = $\frac{\sum_{i=t-29}^t C_i}{30}$
	v2	Weighted moving average (30) = $\frac{\sum_{i=1}^{30} C_{t-i+1} W_{t-i+1}}{\sum_{i=1}^{30} W_{t-i+1}}$
	v3	Commodity channel index (14) = $\frac{TP_t - MATP(14)_t}{\alpha D(14)_t}$, where $TP_t = \frac{H_t + L_t + C_t}{3}$, $\alpha = 0.015$, $MATP(14)_t = \frac{1}{14} \sum_{i=1}^{14} TP_{t-i+1}$, $D(14)_t = \frac{1}{14} \sum_{i=1}^{14} TP_{t-i+1} - MATP(14)_t $
Mean reversion	v4	Fast stochastic oscillator (14) = $K(14)_t = \frac{2}{3} * K(14)_{t-1} + \frac{1}{3} * RSV(14)_t$
	v5	Slow stochastic oscillator (14) = $D(14)_t = \frac{2}{3} * D(14)_{t-1} + \frac{1}{3} * K(14)_t$
Relative strength	v6	Relative strength index (14) = $\frac{UC(14)_t}{UC(14)_t + DC(14)_t} * 100$, where $UC(14)_t = \sum_{i=1}^{14} \max\{0, C_{t-i+1} - C_{t-i}\}$, C_t is the close price at t
	v7	Williams % R (14) = $\frac{HP_{14} - C_t}{HP_{14} - LP_{14}} * 100$, where HP_{14} is the highest price in previous 14 days, and LP_{14} is the lowest price in previous 14 days
Volume	v8	On balance volume = $OBV_t = OBV_{t-1} + \begin{cases} V_t & C_t > C_{t-1} \\ 0 & C_t = C_{t-1} \\ -V_t & C_t < C_{t-1} \end{cases}$
	v9	Money flow index (14) = $\frac{100}{1 + MR(14)_t}$, where $MR(14)$ is the ratio of positive summation of cash flow in previous 14 days and negative summation of cash flow in previous 14 days
Momentum	v10	Moving average convergence divergence (12, 26, 9) = $(DIF - DEA) * 2$, $DIF = EMA_{12} - EMA_{26}$, $DEA = EMA(DIF, 9)$
	v11	Rate of change percentage (10) = $\frac{C_t - C_{t-10}}{C_{t-10}}$
	v12	Momentum (10) = $\frac{C_t}{C_{t-10}} * 100$
	v13	Rate of change (14) = $\frac{C_t - C_{t-14}}{C_{t-14}} * 100$
	v14	Chande momentum oscillator (14) = $\frac{H(14)L(14)}{H(14) + L(14)} * 100$, where $H(14)$ is the sum of higher closes over 14 days, and $L(14)$ is the sum of lower closes over 14 days

Table 2
Financial data as predictor attributes.

Attribute	Variable	Description
Relevant share price rationality	v15	Book value per share = (total equity - preferred equity)/total shares outstanding
	v16	Cash flow per share = (cash flow - preferred dividends)/shares outstanding
	v17	Retention ratio = (net income - dividends)/net income
Profitability	v18	Return on equity(after tax) = net income after tax/shareholders' equity
	v19	Earnings per share = (net income preferred dividends)/shares outstanding
	v20	Net profit margin = net income after tax/net sales
	v21	Gross profit margin = gross profit/net sales
	v22	Net profits = total revenue - total expenses
	v23	Business income measures the amount of gain (in monetary value or in kind) earned from a sale of a service and/or product after deducting all incidental expenses incurred by the business.
	v24	Operating earnings per share = operating income/shares outstanding
Efficiency	v25	Receivable turnover ratio = net credit sales/average accounts receivable
	v26	Receivable turnover (days) measures how many days per year averagely needed by a company to collect its receivables
	v27	Current assets turnover ratio = sales revenue/average current assets
	v28	Current assets turnover (days) measures firm's ability of generating sales through its current assets (cash, inventory, accounts receivable, etc.)
Growth	v29	Operating revenue growth (%) = (operating revenue(t)-operating revenue(t-1))/operating revenue(t)
	v30	Net profit growth (%) = (net profit(t)-net profit(t-1))/net profit(t)
	v31	Net assets growth (%) = (net assets(t)-net assets(t-1))/net assets(t)
	v32	Total assets growth (%) = (total assets(t)- total assets(t-1))/total assets(t)
	v33	Earnings per share growth (%) = (earnings per share(t)- earnings per share(t-1))/earnings per share(t)
Leverage	v34	Current ratio = current assets/current liabilities
	v35	Quick ratio = quick assets/current liabilities
	v36	Cash ratio = (cash + marketable securities)/current liabilities
	v37	Shareholder equity ratio = total shareholder equity/total assets
Liquidity	v38	Operating cash flow to net sales ratio = operating cash flow/net sales
	v39	Operating cash flow to total assets ratio = operating cash flow/total assets
	v40	Operating cash flow to net revenue ratio = operating cash flow/net revenue
	v41	Operating cash flow to net income ratio = operating cash flow/net income
	v42	EBITDA to interest expense ratio = EBITDA/interest expense

indicators are utilized to extract the company's financial information over a long period. Apart from technical indicators and financial ratios, it is essential to use lagged returns as input indicators, according to [Huck \(2019\)](#), [Paiva et al. \(2019\)](#), and [Paiva et al. \(2019\)](#).

Since the US data are downloaded from the WIND database, the exact values of several fundamental variables in [Table 2](#) cannot be determined. Therefore, similar ratios are used as substitutes for the

missing financial information: Business income is replaced by the ratio of operating profit to total profit. The operating earnings per share is replaced by the annual return on invested capital. The current asset turnover (days) is replaced by the operating cycle. The operating cash flow to total assets ratio and the EBITDA to interest expense ratio are replaced by the operating cash flow to operating profit ratio and the cash flow coverage ratio, respectively.

3.3. Process of optimal portfolio formation

The MVF model can be implemented in three steps, that is, through the pairs trading portfolio construction, pairs trading portfolio's return prediction, and pairs trading portfolio allocation, as indicated in the overall flowchart shown in Fig. 2. The aim of the first step is to construct stationary pairs trading portfolios using the Engle–Granger cointegration test and the Johansen test.

In the second step, the stationary pairs portfolios are the target of machine learning models. Data for the first 6 years' (75% of the whole data) are applied to train the A-LSTM network, whereas data for the last 2 years are used for testing the out-of-sample performance in terms of model accuracy and trading performance. The sliding window process is applied to update the parameters of machine learning models day by day. The first two steps confirm that the selected pairs trading portfolios are stationary and highly predictable.

In the last step, only the stationary pairs portfolios with a high prediction accuracy are selected to construct MV portfolios. The weights of the MV portfolio are changed month by month. In the current month, these weights are calculated according to the predicted MV portfolio return using the historical return of each stock pair in the previous month.

4. Experimental results and analysis

A comprehensive evaluation is conducted to analyze the robustness of the proposed model from the perspective of the parameters set for modeling portfolios (holding period and trading threshold), machine learning model selection, and sample selection bias. The first two sections describe the model and trading performance of the stationary portfolios and their corresponding MV portfolios over different holding periods. In the following, the impact of the trading threshold on the proposed MV portfolio is investigated, and the trading performance is evaluated under two cases: without and with the trading threshold. In addition, the trading performance of the proposed MV model is compared with that of two benchmarks that use machine learning models, such as SVM and random forest, and an alternative investment allocation method, such as the naive diversification (1/N) method. Last, the model accuracy and the trading performance of the proposed model are analyzed using US data.

4.1. Pairs trading spread return prediction

The fundamental question in this study is whether stationary analysis improves the return predictability of the A-LSTM network. To answer it, the performance of the A-LSTM model in predicting the return of stationary portfolios (or spread) is compared with that of nonstationary and un-cointegrated stocks.

The performance measures to evaluate the effectiveness of forecasting the return of spread according to Ma et al. (2021) are the mean squared error (MSE), the mean absolute error (MAE), the model accuracy H_R , H_{R+} , and H_{R-} , which are defined as follows.

$$MSE = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2 \quad (19)$$

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t| \quad (20)$$

$$H_R = \frac{\text{Count}_{t=1}^n (y_t \hat{y}_t > 0)}{\text{Count}_{t=1}^n (y_t \hat{y}_t \neq 0)} \quad (21)$$

$$H_{R+} = \frac{\text{Count}_{t=1}^n (y_t > 0, \hat{y}_t > 0)}{\text{Count}_{t=1}^n (\hat{y}_t > 0)} \quad (22)$$

$$H_{R-} = \frac{\text{Count}_{t=1}^n (y_t < 0, \hat{y}_t < 0)}{\text{Count}_{t=1}^n (\hat{y}_t < 0)} \quad (23)$$

where $\text{Count}(\cdot)$ is the counting of occurrences of its argument.

Table 3 presents the model performance for the selected pairs of cointegrated stocks with respect to MSE , MAE , H_R , H_{R+} , and H_{R-} .

Furthermore, the model performance for the portfolio returns over several holding days – 1 day, 2 days, 3 days, 4 days, 1 week, 2 weeks, 3 weeks, and 1 month – are presented. In this case, the machine models are trained day by day by using the daily information of features and predicting the h-day-ahead returns. The nonstationary and un-cointegrated stocks with the corresponding eight prediction horizons are used for comparison; the corresponding results are demonstrated in Panel A, and the prediction results of the A-LSTM model with cointegrated stocks are presented in Panel B.

Four findings can be observed in Table 3. First, based on the result of daily return forecasting, the predictive accuracies of un-cointegrated stocks, that is, the values of H_R , H_{R+} , and H_{R-} , are similar to the findings of Ma et al. (2021) and their reference papers. Second, the predictive accuracies of the stationary portfolio in terms of MSE , MAE , H_R , H_{R+} , and H_{R-} are improved compared with the un-cointegrated stocks. These two findings demonstrate the advantage of combining stationary analysis and the A-LSTM network to predict the daily returns. Third, the prediction performance of the A-LSTM network is significantly improved when the holding period (or prediction horizon) increases and the standard deviations of all measurements do not change too much. This result is true both for individual stocks and spreads. It implies the return predictability does not vanish in the long term within 1, 2 and 3 weeks and 1 month. Last, the return accuracy for upward (H_{R+}) and downward return movement (H_{R-}) are both similar to the total return accuracy (H_R) under different holding periods ($H = 1, 2, \dots, 20$). This result is true both for individual stocks and spreads. It implies the proposed model can predict the return accurately under both upward and downward price movement scenarios.

4.2. Trading performance for MVF

To evaluate the trading performance of the spread prediction-based portfolio, five types of performance – the annualized excess return (AER), the Sharpe ratio (SAR), the Sortino ratio (SOR), the maximum drawdown (MD), and the turnover ratio (TR) – as defined in Eq. (24) to Eq. (27) are examined. The SAR and the SOR are measures of the risk-adjusted return, which are utilized to analyze not only how high the risk premium is but also how small the variation in the return rate is. The AER and the MD evaluate the return and risk of the portfolio's monthly log returns. The turnover rate reflects the change of portfolio weights. A low turnover rate indicates that the portfolio investment requires a lower transaction cost.

$$\text{Sharpe ratio} = \frac{E[r_{i,t}] - r_f}{\text{std}(r_{i,t})} \quad (24)$$

$$\text{Sortino ratio} = \frac{E[r_{i,t}] - r_f}{\text{std}(\min\{r_{i,t}, 0\})} \quad (25)$$

$$\text{Maximum drawdown} = \max_{t < s} \frac{\text{Netv}_t - \text{Netv}_s}{\text{Netv}_t} \quad (26)$$

$$\text{Turnover rate} = \sum_{t=1}^n |w_{i,t} - w_{i,t-1}| \quad (27)$$

where $E[r_{i,t}] - r_f$ is the annualized monthly excess return. $\text{std}(r_{i,t})$ is the standard deviation of the monthly log return. r_f is the monthly risk-free interest rate. $\text{std}(\min\{r_{i,t}, 0\})$ measures the downside risk of the monthly log return. Netv_t represents net value at month t , and $w_{i,t}$ represents the weight of stock i in portfolio at month t .

Table 4 presents the trading performance of the mean-variance portfolio by considering the predicted portfolio return in 1 day, 2 days, ..., and 1 month using the AER, the SAR, the SOR, the MD and the TR. The performance of the nonstationary and un-cointegrated stocks are demonstrated in Panel A, whereas the performance results of the MVF model with cointegrated stocks are presented in Panel B.

The results in the table support that regardless of whether un-cointegrated stocks or stationary portfolios are considered, machine learning models can be used to construct portfolios based on statistical arbitrage and can generate positive trading performance in terms of

Table 3
Predictive performance for CSI 300 stocks in terms of forecast horizon.

		1	2	3	4	5	10	15	20
Panel A: Stock									
MSE	mean (10^{-4})	3.47	5.31	5.80	6.53	6.41	6.69	7.61	7.88
	std (10^{-4})	1.95	3.00	3.17	3.61	3.42	3.47	4.07	4.26
MAE	mean (10^{-2})	1.24	1.55	1.65	1.73	1.74	1.84	1.94	2.01
	std (10^{-3})	4.22	5.33	5.58	5.74	5.73	5.70	6.03	6.73
H_R	mean (%)	46.98	66.05	73.98	77.48	79.46	84.33	86.72	88.14
	std (10^{-2})	3.53	0.41	1.59	1.85	0.46	1.74	1.40	1.15
H_{R+}	mean (%)	42.75	62.33	72.28	76.78	78.87	84.97	87.21	89.40
	std (10^{-2})	3.92	3.46	3.97	5.27	3.89	4.06	3.10	1.78
H_{R-}	mean (%)	46.39	64.23	72.35	75.44	77.15	83.24	85.56	86.62
	std (10^{-2})	6.76	4.86	3.19	1.93	3.53	4.42	2.35	2.91
Panel B: Spread									
MSE	mean (10^{-4})	3.21	3.49	5.52	5.29	5.86	5.25	6.63	7.37
	std (10^{-4})	1.87	1.99	2.24	1.75	2.20	2.38	2.95	2.98
MAE	mean (10^{-2})	1.21	1.28	1.68	1.64	1.75	1.64	1.85	1.96
	std (10^{-3})	3.67	4.18	3.15	3.07	3.46	4.38	4.80	5.72
H_R	mean (%)	53.17	68.21	75.75	80.06	81.89	87.74	90.80	92.59
	std (10^{-2})	1.45	0.71	0.77	0.95	1.36	0.40	0.51	1.00
H_{R+}	mean (%)	52.12	67.00	73.74	79.05	81.25	87.06	91.82	93.00
	std (10^{-2})	1.59	1.87	1.52	1.58	1.01	1.83	1.42	1.40
H_{R-}	mean (%)	54.07	68.98	77.29	80.62	82.41	88.22	90.04	92.31
	std (10^{-2})	2.22	1.87	1.70	2.49	1.80	1.88	1.33	1.73

Table 4
Trading performance of the mean-variance portfolio using CSI 300 stocks with and without spread prediction.

		1	2	3	4	5	10	15	20	N_o
Panel A: Stock										
AER	-0.87	1.46	0.26	2.25	0.85	0.82	1.40	0.10	-0.47	
SAR	-0.65	1.25	0.21	1.98	0.98	0.94	1.28	0.07	-0.42	
SOR	-0.75	1.27	0.25	1.85	1.42	1.34	1.67	0.15	-0.54	
MD	3.31	2.86	3.52	2.86	1.73	1.73	2.06	1.79	5.97	
TR(%)	5.20	6.00	5.99	1.06	6.00	5.60	5.20	6.80	1.06	
Panel B: Spread										
AER	5.23	1.51	3.84	11.68	8.41	10.36	11.30	8.83	-2.13	
SAR	4.39	1.51	3.53	12.92	8.70	11.41	11.86	9.31	-2.03	
SOR	8.70	5.03	6.08	28.00	22.94	134.91	87.00	22.59	-2.98	
MD	3.23	1.25	3.40	1.21	1.97	1.52	3.29	2.15	2.48	
TR(%)	8.31	8.94	7.24	8.14	8.13	7.65	6.77	6.48	0.00	

risk and return measures, in line with the results of similar empirical studies (Flori & Regoli, 2021; Huck, 2019). More importantly, the trading performance of the proposed A-LSTM+MVf model on cointegrated stocks is superior to that shown in the literature, which combines machine learning models, MV allocation, and statistical arbitrage from the perspective of the Sharpe and Sortino ratios (Freitas et al., 2009; Krauss et al., 2017; Wang et al., 2020).

A clear advantage of using the A-LSTM+MVf model on stationary portfolios is also shown in Table 4 based on two benchmarks. First, compared with the performance on un-cointegrated stocks, the optimal MVf portfolios with the selected cointegrated spreads have a larger AER, SAR, SOR, and MD under different holding periods. It implies that the return (risk) of the optimal spread-based MVf portfolio is high (low). Meanwhile, the TR for all MVf portfolios that are composed of stocks and spreads is small, which indicates a low transaction cost is required to construct the MVf portfolio. Second, the MVf is a significant improvement compared with the conventional MV model without prediction, since the expected return from the conventional MV portfolio cannot cover the risk-free interest rate return. The performance result for stationary portfolios is stronger than that for un-cointegrated stocks.

Determining the optimal holding period return of the MVf portfolio offers instructive information for investors to construct MVf portfolios and develop profitable trading strategies in the real world. It is found from the perspective of different portfolio holding periods that the performance on holding the pairs trading portfolio for 2–3 weeks is the best compared with the performance on holding it for the other periods.

Specifically, in the A-LSTM+MVf model, the portfolio with $H = 10$ and 15 not only has a high SAR, ranging from 11.41 to 11.86, and an SOR ranging from 87.00 to 134.91, but also has a lower TR that ranges from 6.77 to 7.65.

4.3. Impact of the trading threshold on portfolio rebalancing

So far, the analysis of the model performance and the trading performance of the proposed model does not consider the trading threshold. In fact, rational investors have the motivation to reallocate the MVf portfolio every month only when the predicted portfolio return during the next 1 day, 2 days, ..., 1 month satisfies their expectations, that is, it is larger than 0 or the interest rate. With the effect of the constraint of the trading threshold, the profitability of the portfolio with predicted spreads is reexamined by comparing the results when there is no threshold, as shown in Table 4, with the results when there is a threshold, as shown in Table 5.

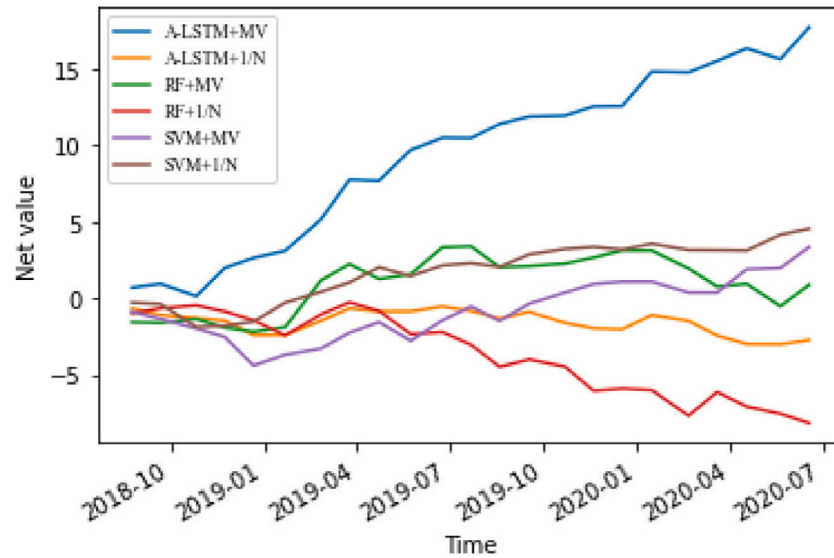
Table 5 considers two cases for triggering portfolio allocation: changing the weight of the portfolio when the predicted portfolio return exceeds 0 (Panel A) or the interest rate (Panel B).

Two main results are found. First, although the trading strategies with the zero or the interest rate constraint are still profitable and robust, either the constraint of zero or of the interest rate reduces the performance of AER, SAR, SOR, MD, and TR. Second, when the threshold changes from 0 to the monthly interest rate, when the prediction horizon exceeds 4 days, whether the trading threshold (either 0 or interest rate) is used or not does not seem to matter in the

Table 5

Trading performance for mean-variance portfolio using CSI 300 stocks with regard to threshold.

	1	2	3	4	5	10	15	20
Panel A: $M = 0$								
AER	4.68	0.73	4.63	11.52	8.86	8.98	11.10	8.74
SAR	5.40	0.72	4.79	13.76	9.07	8.26	11.54	9.04
SOR	10.18	2.03	13.52	61.12	24.27	10.95	75.85	21.84
MD	4.19	1.43	2.76	1.21	1.97	1.60	3.29	2.15
TR(%)	7.72	8.00	7.60	8.17	7.20	7.25	7.04	6.66
Panel B: $M = \text{interest rate}$								
AER	0.17	0.45	3.56	11.32	7.27	8.86	11.04	8.61
SAR	1.00	0.45	3.78	13.27	6.75	8.13	11.42	9.03
SOR	NaN	1.51	9.39	72.05	16.39	11.18	73.63	20.76
MD	1.47	1.43	2.71	1.21	1.97	1.60	3.29	2.15
TR(%)	0.00	7.58	7.65	8.14	7.52	7.25	7.11	5.78

**Fig. 3.** Net value of different models..**Table 6**

Trading performance of other models using CSI 300 stocks.

	A-LSTM+1/N	RF+MV	RF+1/N	SVM+MV	SVM+1/N
AER	-1.39	0.42	-4.10	1.65	2.25
SAR	-2.57	0.38	-4.62	1.79	3.69
SOR	-4.67	0.63	-8.37	3.18	4.89
MD	2.03	3.23	3.99	2.19	1.47

evaluation of the trading performance. Both these results imply that the trading threshold has a negative impact on the proposed MVF portfolio, especially for investors who focus on short-term portfolio returns. The explanation for this finding is that if the investors only trade when the predicted portfolio return exceeds 0 or the interest rate using the proposed model, they may lose some profitable trading opportunities.

4.4. Model variations and performance results

This section discusses the trading performance by considering variations of the predictive machine learning model and the asset allocation method. Random forest and SVM are considered two alternative machine learning models to A-LSTM networks, and the 1/N portfolio construction method is considered an alternative asset allocation method to the MVF model. The trading performance of the variations of possible portfolios with 20-day-ahead machine learning predictions over the last two years is summarized in Table 6.

First, three MVF models with machine learning model forecasts are compared. As shown in Tables 5 and 6, A-LSTM+MVF has the highest

AER, SAR, and SOR, and the lowest MD compared with RF+MV and SVM+MV. In addition, Fig. 3 shows the superiority of A-LSTM+MVF to RF+MV and SVM+MV in terms of net value. Therefore, A-LSTM+MVF outperforms RF+MV and SVM+MV.

Second, two portfolio allocation methods, MVF and 1/N, are compared. Tables 5 and 6 show that A-LSTM+MVF has highly improved A-LSTM+1/N, in terms of AER, SAR, and SOR, because A-LSTM+1/N has a negative Sharpe/Sortino ratio. It implies that the expected return from the A-LSTM+1/N portfolio is less than the risk-free return. In this case, rational investors who can either buy treasury bonds or equally invest in stock market prefer to buy treasury bonds. Since the AER for A-LSTM+MVF is positive for all spreads, A-LSTM+MVF performs better than A-LSTM+1/N and treasury bond investments. This is the same for RF+MV. Moreover, SVM+1/N slightly outperforms SVM+MV in terms of AER, SAR, SOR, and MD. From Fig. 3, the net value of A-LSTM+MVF and of SVM+MV are significantly higher than that of A-LSTM+1/N and of SVM+1/N, respectively. Moreover, SVM+1/N slightly outperforms SVM+MV in terms of net value. Since two out of three machine learning models strongly support that the MVF investment portfolio outperforms the 1/N investment portfolio, and MVF outperforms 1/N in terms of all trading performance measures and net value.

Third, six models composed by using all possible combinations of the three MVF models and the two portfolio allocation methods are compared in Table 6 and Fig. 3. The A-LSTM+MVF model is superior to the other five models in terms of AER, SAR, SOR, MD, and the net value of the portfolio over the last two years.

Table 7

Predictive performance for spreads of US stocks in terms of forecast horizon.

		1	2	3	4	5	10	15	20
MSE	mean (10^{-4})	6.53	9.14	11.74	13.09	14.20	16.82	24.14	33.40
	std (10^{-4})	4.32	6.31	8.37	9.11	10.58	14.00	35.28	64.50
MAE	mean (10^{-2})	1.64	1.98	2.20	2.33	2.45	2.64	2.96	3.29
	std (10^{-3})	5.06	6.62	7.17	7.33	8.44	9.30	1.47	2.00
H_R	mean (%)	54.35	66.36	71.61	75.92	78.52	85.50	86.88	88.52
	std (10^{-2})	1.08	1.62	2.29	2.33	1.64	2.01	2.92	3.38
H_{R+}	mean (%)	55.10	67.74	73.01	77.55	80.35	87.98	89.69	91.72
	std (10^{-2})	2.35	1.90	2.73	1.88	2.82	2.93	3.94	3.96
H_{R-}	mean (%)	53.69	64.74	70.03	74.17	76.50	82.68	83.77	85.40
	std (10^{-2})	2.17	4.27	4.16	4.87	4.28	5.75	7.15	8.54

Table 8

Trading performance for mean–variance portfolio based on stationary portfolios of S&P 500 stocks with and without threshold.

	1	2	3	4	5	10	15	20
Panel A: No threshold constraint								
AER	6.32	15.41	36.14	49.03	37.73	85.55	133.81	150.58
SAR	1.16	1.42	2.42	2.76	3.05	2.27	3.48	2.77
SOR	1.42	2.31	4.01	4.83	11.01	3.23	2.70	4.77
MD	1.93	1.77	1.67	1.41	1.37	1.92	1.81	1.71
TR(%)	0.20	1.71	1.10	1.37	1.74	1.99	1.16	0.96
Panel B: M = 0								
AER	4.26	18.51	33.49	53.59	37.70	94.67	141.77	160.71
SAR	0.81	1.92	2.22	3.46	3.05	2.76	3.86	3.12
SOR	1.09	2.56	3.51	6.81	11.00	3.49	NaN	4.31
MD	1.95	1.77	1.67	1.40	1.37	1.92	1.81	1.71
TR(%)	0.70	1.71	1.10	1.37	1.74	1.99	1.28	0.96
Panel C: M = interest rate								
AER	2.20	18.46	33.17	53.77	37.70	95.18	142.20	160.95
SAR	0.43	1.94	2.19	3.48	3.05	2.79	3.88	3.12
SOR	0.66	2.51	3.73	6.79	11.00	3.46	NaN	4.31
MD	2.62	1.77	1.67	1.39	1.37	1.92	1.81	1.71
TR(%)	2.00	1.71	1.10	1.37	1.74	1.99	1.29	0.96

In conclusion, the spread selection using the A-LSTM network and the MV forecasting model together guarantees the best trading performance.

4.5. Model performance for US data

This section checks whether the proposed model is robust to using data on stocks in another country. It examines the predictability and profitability of the proposed model using 20 stocks that are randomly selected from the S&P 500. The results are provided in [Tables 7](#) and [8](#).

Three main results for the S&P 500 are consistent with those for the CSI 300 in [Tables 3](#) and [4](#). First, [Table 7](#) shows that the A-LSTM+MVf model on stationary portfolios constructed by including US stock samples performs as well as the model that includes the Chinese stock samples under all holding periods. The result is also true for US stocks that as the holding period increases, the return movements over the holding period can be predicted more accurately.

Second, [Table 8](#) supports the result that in the US stock market, machine learning techniques and statistical arbitrage can be combined to generate positive trading performance in terms of risk and return measures. In particular, according to the Sharpe ratio and the Sortino ratio, the proposed A-LSTM+MVf model using spread prediction outperforms the relevant study using LSTM ([Fischer & Krauss, 2018](#)) based on the daily and weekly return of S&P 500 stocks.

Third, from [Table 8](#), the trading performance under all three scenarios – no threshold, zero threshold, and interest rate threshold – are positive in terms of risk-adjusted return measures. Thus, the trading performance of the stationary portfolios in the US are not sensitive to trading thresholds.

These three findings all confirm that the high accuracy and trading performance revealed through combining stationary analysis and the A-LSTM+MVf model hold true for the cointegrated S&P 500 stocks.

5. Conclusion

This study proposed a data-driven model that can optimize trading investments. Specifically, it developed an investment decision model based on the A-LSTM network to select the stationary portfolios on cointegrated assets and allocate them to construct the MV portfolio. This study reveals several important findings.

First, both the classification and the regression accuracy of the A-LSTM network trained with short-term trading signals (technical indicators), long-term trading signals (financial ratios), and lagged returns present significant results. The daily return predictability of the A-LSTM model on stationary portfolios is higher than that of nonstationary and un-cointegrated stocks. This result does not vanish in the long term within 1, 2, and 3 weeks, and 1 month.

Second, the monthly portfolio performance demonstrates that integrating pairs trading and the A-LSTM network with MV portfolio construction is profitable. The proposed portfolio outperforms the conventional MV portfolio under different holding periods.

Third, the cumulative return performance without transaction costs shows a significant result. After comparing the outcomes of A-LSTM+MVf against SVM+MVf, RF+MVf, A-LSTM+1/N, SVM+1/N, and RF+1/N, the proposed portfolio is appropriate for stationary time series forecasting, for it beats these early machine learning models and the 1/N investment allocation method by a very clear margin.

Last, the accuracy of the A-LSTM model for predicting the return of the pairs portfolios and the trading performance of the MV portfolio with the A-LSTM based pairs portfolios is analyzed by employing US data. This analysis confirms that the accuracy and trading performance of the combination of the stationary analysis, the A-LSTM network and the MVf model are high for cointegrated S&P 500 stocks as well.

Thus, this study extends the theoretical literature on machine learning and stationary analysis. It is unique in that it improves the predictive accuracy of machine learning models by predicting the stationary

time series portfolio return instead of nonstationary time series stock returns. This study also extends the theoretical literature on pairs trading and MV portfolio analysis. The combination of pairs trading and MV portfolio analysis guarantees the proposed portfolio is a risk-neutral portfolio and is well-diversified for risk-averse investors.

In the present study, the Engle–Granger cointegration test and the Johansen test are utilized to construct bivariate cointegration-based portfolios. However, it is difficult when the sample size for cointegrated stocks is large. Further research about stationary analysis for multiple cointegrated stocks can be conducted in the future. This study also is limited because it only utilizes structure data, such as technical indicators, financial investment information, and lagged returns as input features in order to compare with the benchmark model. In future studies, more exogenous variables, such as news events, investor sentiment, and policy changes will be incorporated into the model. Therefore, more LSTM cells for different exogenous variables and more attention layers are expected to be added to the A-LSTM model.

CRedit authorship contribution statement

Juan Du: Software, Methodology, Validation, Resource, Data curation, Writing, Fund acquisition, Formal analysis, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We acknowledge the financial support from Shandong Provincial Natural Science Foundation (Grant Reference Number: ZR2021QG065).

References

- Alonso-Monsalve, S., Suárez-Cetrulo, A. L., Cervantes, A., & Quintana, D. (2020). Convolution on neural networks for high-frequency trend prediction of cryptocurrency exchange rates using technical indicators. *Expert Systems with Applications*, 149, Article 113250.
- Barberis, N., Shleifer, A., & Wurgler, J. (2005). Comovement. *Journal of Financial Economics*, 75(2), 283–317.
- Ceria, S., & Stubbs, R. A. (2006). Incorporating estimation errors into portfolio selection: Robust portfolio construction. *Journal of Asset Management*, 7(2), 109–127.
- Chen, Y., & Hao, Y. (2017). A feature weighted support vector machine and K-nearest neighbor algorithm for stock market indices prediction. *Expert Systems with Applications*, 80, 340–355.
- Chen, W., Zhang, H., Mehrlawat, M. K., & Jia, L. (2021). Mean-variance portfolio optimization using machine learning-based stock price prediction. *Applied Soft Computing*, 100, Article 106943.
- Chiu, M. C., & Wong, H. Y. (2011). Mean-variance portfolio selection of cointegrated assets. *Journal of Economic Dynamics and Control*, 35(8), 1369–1385.
- Chiu, M. C., & Wong, H. Y. (2012). Mean-variance asset-liability management: Cointegrated assets and insurance liability. *European Journal of Operational Research*, 223(3), 785–793.
- Chong, E., Han, C., & Park, F. C. (2017). Deep learning networks for stock market analysis and prediction: Methodology, data representations, and case studies. *Expert Systems with Applications*, 83, 187–205.
- Chopra, V. K., Hensel, C. R., & Turner, A. L. (1993). Massaging mean-variance inputs: returns from alternative global investment strategies in the 1980s. *Management Science*, 39(7), 845–855.
- Conrad, J., & Kaul, G. (1998). An anatomy of trading strategies. *Review of Financial Studies*, 11(3), 489–519.
- Fama, E. F., & French, K. R. (2008). Average returns, B/M, and share issues. *The Journal of Finance*, 63(6), 2971–2995.
- Fischer, T., & Krauss, C. (2018). Deep learning with long short-term memory networks for financial market predictions. *European Journal of Operational Research*, 270(2), 654–669.
- Fliege, J., & Werner, R. (2014). Robust multiobjective optimization & applications in portfolio optimization. *European Journal of Operational Research*, 234(2), 422–433.
- Flori, A., & Regoli, D. (2021). Revealing pairs-trading opportunities with long short-term memory networks. *European Journal of Operational Research*, 295(2), 772–791.
- Freitas, F. D., De Souza, A. F., & de Almeida, A. R. (2009). Prediction-based portfolio optimization model using neural networks. *Neurocomputing*, 72(10–12), 2155–2170.
- Goldfarb, D., & Iyengar, G. (2003). Robust portfolio selection problems. *Mathematics of Operations Research*, 28(1), 1–38.
- Green, T. C., & Hwang, B. H. (2009). Price-based return comovement. *Journal of Financial Economics*, 93(1), 37–50.
- Greenwood, R. (2008). Excess comovement of stock returns: Evidence from cross-sectional variation in Nikkei 225 weights. *Review of Financial Studies*, 21(3), 1153–1186.
- Gregory, C., Darby-Dowman, K., & Mitra, G. (2011). Robust optimization and portfolio selection: The cost of robustness. *European Journal of Operational Research*, 212(2), 417–428.
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *Review of Financial Studies*, 33(5), 2223–2273.
- Han, Y., Yang, K., & Zhou, G. (2013). A new anomaly: The cross-sectional profitability of technical analysis. *Journal of Financial and Quantitative Analysis*, 48(5), 1433–1461.
- Heaton, J. B., Polson, N. G., & Witte, J. H. (2017). Deep learning for finance: Deep portfolios. *Applied Stochastic Models in Business and Industry*, 33(1), 3–12.
- Hilliard, J., & Zhang, H. (2015). Size and price-to-book effects: Evidence from the Chinese stock markets. *Pacific-Basin Finance Journal*, 32, 40–55.
- Hjalmarsson, E. (2010). Predicting global stock returns. *Journal of Financial and Quantitative Analysis*, 49–80.
- Huang, S., Huang, Y., & Lin, T. C. (2019). Attention allocation and return co-movement: Evidence from repeated natural experiments. *Journal of Financial Economics*, 132(2), 369–383.
- Huck, N. (2010). Pairs trading and outranking: The multi-step-ahead forecasting case. *European Journal of Operational Research*, 207(3), 1702–1716.
- Huck, N. (2019). Large data sets and machine learning: Applications to statistical arbitrage. *European Journal of Operational Research*, 278(1), 330–342.
- Jiang, W. (2021). Applications of deep learning in stock market prediction: recent progress. *Expert Systems with Applications*, 184, Article 115537.
- Kara, Y., Boyacioglu, M. A., & Baykan, Ö. K. (2011). Predicting direction of stock price index movement using artificial neural networks and support vector machines: The sample of the Istanbul stock exchange. *Expert Systems with Applications*, 38(5), 5311–5319.
- Kim, Y., & Enke, D. (2016). Developing a rule change trading system for the futures market using rough set analysis. *Expert Systems with Applications*, 59, 165–173.
- Krauss, C., Do, X. A., & Huck, N. (2017). Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500. *European Journal of Operational Research*, 259(2), 689–702.
- Lewellen, J. (2004). Predicting returns with financial ratios. *Journal of Financial Economics*, 74(2), 209–235.
- Long, W., Lu, Z., & Cui, L. (2019). Deep learning-based feature engineering for stock price movement prediction. *Knowledge-Based Systems*, 164, 163–173.
- Ma, Y., Han, R., & Wang, W. (2021). Portfolio optimization with return prediction using deep learning and machine learning. *Expert Systems with Applications*, 165, Article 113973.
- Macedo, L. L., Godinho, P., & Alves, M. J. (2017). Mean-semivariance portfolio optimization with multiobjective evolutionary algorithms and technical analysis rules. *Expert Systems with Applications*, 79, 33–43.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7, 77–91.
- Nam, K., Washer, K. M., & Chu, Q. C. (2005). Asymmetric return dynamics and technical trading strategies. *Journal of Banking & Finance*, 29(2), 391–418.
- Neely, C. J., Rapach, D. E., Tu, J., & Zhou, G. (2014). Forecasting the equity risk premium: the role of technical indicators. *Management Science*, 60(7), 1772–1791.
- Paiva, F. D., Cardoso, R. T. N., Hanaoka, G. P., & Duarte, W. M. (2019). Decision-making for financial trading: A fusion approach of machine learning and portfolio selection. *Expert Systems with Applications*, 115, 635–655.
- Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015a). Predicting stock and stock price index movement using trend deterministic data preparation and machine learning techniques. *Expert Systems with Applications*, 42(1), 259–268.
- Patel, J., Shah, S., Thakkar, P., & Kotecha, K. (2015b). Predicting stock market index using fusion of machine learning techniques. *Expert Systems with Applications*, 42(4), 2162–2172.
- Silva, A., Neves, R., & Horta, N. (2015). A hybrid approach to portfolio composition based on fundamental and technical indicators. *Expert Systems with Applications*, 42(4), 2036–2048.
- Vidyamurthy, G. (2004). *Pairs trading: Quantitative methods and analysis*, vol. 217. John Wiley & Sons.
- Wang, W., Li, W., Zhang, N., & Liu, K. (2020). Portfolio formation with preselection using deep learning from long-term financial data. *Expert Systems with Applications*, 143, Article 113042.

- Weng, B., Lu, L., Wang, X., Megahed, F. M., & Martinez, W. (2018). Predicting short-term stock prices using ensemble methods and online data sources. *Expert Systems with Applications*, 112, 258–273.
- Wu, W., Chen, J., Yang, Z., & Tindall, M. L. (2021). A cross-sectional machine learning approach for hedge fund return prediction and selection. *Management Science*, 67(7), 4577–4601.
- Yu, J. R., Chiou, W.-J. P., Lee, W. Y., & Lin, S. J. (2020). Portfolio models with return forecasting and transaction costs. *International Review of Economics & Finance*, 66, 118–130.
- Zymler, S., Rustem, B., & Kuhn, D. (2011). Robust portfolio optimization with derivative insurance guarantees. *European Journal of Operational Research*, 210(2), 410–424.